

Document 7 – Data Model & Synthetic Dataset Design

Smart Destination Recommender for Sustainable Tourism

1. Purpose

The Smart Destination Recommender relies on data to generate personalized travel recommendations while balancing tourist demand, destination sustainability, and traveler preferences.

As part of the MVP, a structured dataset has been designed to simulate realistic tourism scenarios across Spain. The dataset enables the recommendation engine to evaluate destination characteristics, traveler interests, congestion levels, and sustainability indicators in order to produce explainable recommendations.

The objective is to demonstrate how Artificial Intelligence can help TUI improve customer experience while supporting sustainable tourism strategies and ESG objectives.

2. Dataset Overview

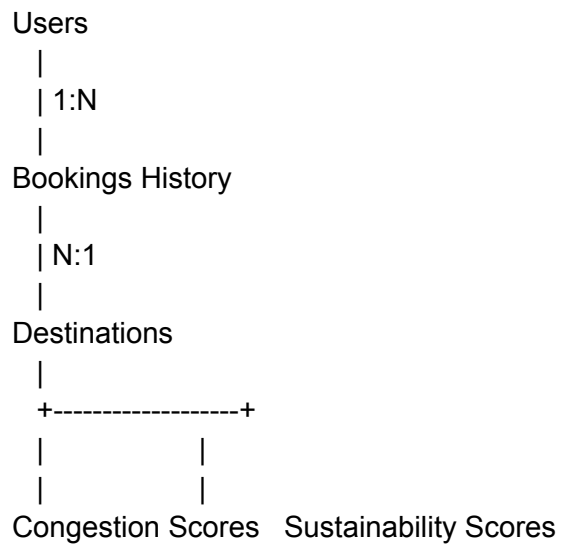
The MVP dataset is organized into five logical domains:

Domain	Description
Destinations	Tourist destinations available for recommendation
Users	Simulated traveler profiles
Bookings History	Historical traveler interactions
Congestion Scores	Seasonal destination demand indicators
Sustainability Scores	ESG-related destination indicators

Together, these datasets provide the minimum viable information required to support recommendation generation, explainability, and future machine learning enhancements.

3. Data Model

The recommendation engine is based on a simple relational model connecting travelers, destinations, and destination performance indicators.



Entity Description

Users

Represents traveler profiles and travel preferences.

Destinations

Represents tourist destinations available within Spain.

Bookings History

Stores simulated traveler interactions and ratings.

Congestion Scores

Stores seasonal tourism demand indicators.

Sustainability Scores

Stores environmental, mobility, and local economic sustainability metrics.

4. Dataset Structure

4.1 Destinations Dataset

Stores destination attributes used by the recommendation engine.

Table: destinations.csv

Field	Type	Description
destination_id	String	Unique destination identifier
city	String	Destination name
region	String	Spanish region
country	String	Country
beach	Integer	Beach attractiveness score (1–5)
culture	Integer	Cultural attractiveness score (1–5)
nature	Integer	Nature attractiveness score (1–5)
nightlife	Integer	Nightlife attractiveness score (1–5)
family_friendly	Integer	Family suitability score (1–5)
avg_price_per_day	Integer	Estimated daily travel cost (€)

Example Destinations

- Mallorca
- Ibiza
- Tenerife

- Barcelona
- Valencia
- Málaga
- Seville
- Granada
- Bilbao
- San Sebastián
- Alicante
- Menorca
- Lanzarote
- Cádiz
- Zaragoza
- Córdoba
- Santiago de Compostela
- Murcia
- Salamanca
- Girona

4.2 Users Dataset

Stores traveler demographic information and travel preferences.

Table: users.csv

Field	Type	Description
user_id	String	Unique user identifier
age	Integer	Traveler age
country	String	Country of origin
beach_pref	Integer	Preference for beach destinations
culture_pref	Integer	Preference for cultural experiences
nature_pref	Integer	Preference for nature activities
nightlife_pref	Integer	Preference for nightlife
family_pref	Integer	Preference for family-oriented travel
sustainability_interest	Integer	Interest in sustainable tourism

The MVP will simulate approximately 100 traveler profiles representing key TUI source markets.

Proposed Distribution

Country	Percentage
Germany	30%
United Kingdom	25%
Netherlands	20%
France	15%
Nordic Countries	10%

4.3 Bookings History Dataset

Represents previous traveler interactions with destinations.

Table: bookings_history.csv

Field	Type	Description
user_id	String	Traveler identifier
destination_id	String	Destination identifier
rating	Integer	Traveler rating (1–5)

This dataset supports future collaborative filtering and recommendation personalization techniques.

4.4 Congestion Scores Dataset

Stores destination crowding levels by season.

Table: congestion_scores.csv

Field	Type	Description
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destination_id	String	Destination identifier
month	String	Month
congestion_score	Integer	Crowding indicator (0–100)

Congestion Scale

Score	Interpretation
0–30	Low congestion
31–60	Medium congestion
61–80	High congestion
81–100	Very high congestion

The congestion score represents estimated tourism pressure and is used to encourage more balanced destination recommendations.

4.5 Sustainability Scores Dataset

Stores ESG-related indicators used by the recommendation engine.

Table: sustainability_scores.csv

Field	Type	Description
destination_id	String	Destination identifier
carbon_score	Integer	Environmental performance score
local_business_score	Integer	Local economy support score
public_transport_score	Integer	Sustainable mobility score
sustainability_score	Integer	Composite sustainability metric

Sustainability Formula

Sustainability Score =
 $0.4 \times \text{Carbon Score} +$
 $0.3 \times \text{Local Business Score} +$
 $0.3 \times \text{Public Transport Score}$

The resulting score provides a simplified sustainability indicator for recommendation ranking.

5. Data Generation and Source Strategy

The MVP combines real destination metadata with synthetic traveler and booking information.

This approach balances realism, privacy, and rapid prototyping while maintaining a clear path toward enterprise deployment.

Real-World Inspired Data

Destination information is based on publicly available tourism information, destination descriptions, and travel industry sources.

Examples include:

- Spanish tourism websites
- Travel booking platforms
- Public tourism statistics
- Destination marketing organizations
- Sustainability and mobility reports

The selected destinations correspond to real Spanish tourist destinations commonly offered by TUI and other travel operators.

Destination attributes such as beach appeal, cultural attractions, nature experiences, and nightlife are derived from publicly available destination information and expert assessment.

Synthetic Data

The following datasets are synthetically generated:

- Traveler profiles
- Historical bookings

- User ratings
- Recommendation interactions

No customer information or personally identifiable information (PII) is used.

This ensures compliance with privacy requirements while enabling realistic recommendation scenarios.

Congestion and Sustainability Indicators

For the MVP, congestion and sustainability scores are modeled using realistic tourism patterns and ESG criteria.

The values are designed to represent relative destination performance rather than official measurements.

In a production implementation, these indicators could be calculated from:

- Tourism arrival statistics
 - Hotel occupancy rates
 - Mobility and transportation datasets
 - Environmental sustainability indicators
 - Local economic impact metrics
 - Seasonal demand forecasting models
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Production Evolution

The proposed architecture is designed to replace synthetic inputs with operational data feeds without requiring changes to the recommendation engine.

Future versions could integrate data from:

- TUI booking platforms
- Destination management systems
- Tourism authorities
- Sustainability reporting providers
- Mobility and transportation platforms

This enables a smooth transition from MVP validation to enterprise-scale deployment.

6. Dataset Statistics

The MVP dataset is intentionally limited in size to support rapid development and demonstration.

Dataset	Estimated Records
Destinations	20
Users	100
Bookings History	500
Congestion Records	240
Sustainability Records	20

Despite its limited size, the dataset is sufficient to demonstrate recommendation logic, explainability, and sustainability-aware decision making.

7. AI Features

The recommendation engine combines traveler preferences with destination characteristics and sustainability metrics.

Key features include:

- Preference Match Score
- Sustainability Score
- Congestion Score
- Historical Rating Signals

Preference Match

Measures similarity between traveler interests and destination attributes.

Sustainability Score

Measures environmental and social performance.

Congestion Score

Measures tourism pressure and overcrowding.

Historical Ratings

Captures previous traveler satisfaction.

8. Recommendation Scoring Logic

The MVP uses a weighted recommendation formula.

Final Score =
 $0.50 \times \text{Preference Match}$
+
 $0.25 \times \text{Sustainability Score}$
+
 $0.25 \times (100 - \text{Congestion Score})$

This approach rewards destinations that:

- Match traveler preferences
- Demonstrate stronger sustainability performance
- Experience lower tourism congestion

The formula can later be replaced by machine learning models while preserving the same feature structure.

9. Expected Business Outcome

The dataset enables the Smart Destination Recommender to generate personalized and explainable travel recommendations.

The solution demonstrates how TUI can:

- Improve traveler satisfaction
- Reduce pressure on overcrowded destinations
- Promote sustainable tourism practices

- Increase visibility of alternative destinations
- Support ESG objectives through AI-driven decision making
- Create a foundation for future AI-powered recommendation services

The dataset serves as the foundational data layer of the MVP and is designed to evolve into a production-grade recommendation platform through integration with operational tourism and booking data sources.